ECON 5030 Midterm Study Guide

A 4-hour plan, including Cournot from scratch

Built from the 10/22 review, the course notes (Ch. 1–3, Ch. 5), PS1–PS3, and last year's midterm

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1 What is on the exam, and the 4-hour plan

What the professor said (10/22 review)

- **Preferences:** define complete and transitive, and know the rationality axioms and when a utility function exists. "Easier than PS1, more definitions than proofs."
- **Expected utility:** what EU is and what extra assumption it needs (*independence*).
- **Normal form:** "will 100% ask." Know the three parts (players, strategies, payoffs), and how to build one for simultaneous and sequential games.
- **Solution concepts:** dominant strategies, rationalizability / iterated deletion, Nash equilibrium (know the definition), and mixed NE.
- **Efficiency and equity:** NE is generally neither.
- **Market games:** CS, PS, TS, DWL; what blocks pure exchange; what production adds.
- **Cournot oligopoly:** "the biggest of the problems on the exam," "a Cournot example from the problem set will probably be on the exam."
- **Not on the midterm:** neoclassical demand (UMAX/Lagrangian) and Bertrand.

Last year's midterm (transcribed in `ECON5030_Midterm.pdf`) had four sections: (1) Preferences, (2) Game Theory (a sequential game written out in normal form, plus a money game with "Case 1/2/3" preferences), (3) Conceptual, and (4) General equilibrium. Expect the same shape, with Section 4 replaced by Cournot.

The 4-hour schedule

Time	Do this	Guide section
0:00–0:25	Preferences and utility: memorize the definitions, then do the three "is it complete / transitive / representable?" drills.	§2
0:25–0:45	Expected utility and the independence axiom. Write out the answer to "why exactly independence?" from memory once.	§3
0:45–1:45	Game theory: normal form, sequential strategies, IDSDS, NE, SPE, mixed NE. Work every example with pencil, especially the 2-step sequential game recipe.	§4
1:45–1:55	**Break.**	
1:55–3:10	**Cournot from scratch.** Read the recipe, then solve PS3 #1 and #2 yourself before looking at the solutions. Draw the BR graph and the welfare graph.	§5, §6
3:10–3:25	Market games and welfare: CS/PS/TS/DWL, equity, pure exchange vs. production.	§7
3:25–4:00	**Timed practice:** Sample Midterm 2, Sections 2.2 and 4 (25 min), then grade with its key (10 min). Finish with the final checklist (§9).	§8, §9

Exam tip

Grading is "long, just show what you've learnt." On every question: **state the definition first**, then apply it. Definitions are free points, and the professor explicitly rewards them.

2 Preferences and utility (25 min)

Core definitions (memorize word for word)

- **Complete:** $(\succsim, S)$ is complete if for all $x, y \in S$, either $x \succsim y$ or $y \succsim x$ (or both).
- **Transitive:** $(\succsim, S)$ is transitive if for all $x, y, z \in S$, $x \succsim y$ and $y \succsim z$ imply $x \succsim z$.
- **Rational** = complete + transitive.
- **Strict preference / indifference:** $x \succ y$ iff $x \succsim y$ and not $y \succsim x$; $x \sim y$ iff $x \succsim y$ and $y \succsim x$.
- **Representation:** $u : S \to \mathbb{R}$ represents $(\succsim, S)$ if for all x, y: $x \succsim y \iff u(x) \geq u(y)$.
- **Continuous:** for any sequences with $x^n \succsim y^n$, $x^n \to x$, $y^n \to y$, we have $x \succsim y$. (No "jumps" at limits.)

The existence results (the story the professor wants):

1. If S is **finite** and $\succsim$ is complete and transitive, then an optimal choice exists and a utility function exists. (Rank the indifference classes $1, 2, \ldots, K$ from worst to best and set $u(x) =$ the rank of x's class.)
2. If $X \subseteq \mathbb{R}^L$ is **infinite**, complete + transitive is *not* enough. You also need **continuity**. Complete + transitive + continuous $\Rightarrow$ a continuous utility function exists.
3. Optimal choices $= \arg\max u$.
4. Utility is **ordinal**: any strictly increasing transformation of u represents the same preferences.

The standard counterexample, lexicographic preferences on $[0,1]^2$: $x \succ y$ if $x_1 > y_1$, or $x_1 = y_1$ and $x_2 > y_2$. These are complete and transitive but *not continuous*: $(1/n, 0) \succ (0,1)$ for every n, but the limit $(0,0) \prec (0,1)$. No utility function exists. (Intuition: each x_1 needs its own disjoint utility interval, and there are uncountably many of them but only countably many rationals to tag them with.)

Two facts that answer "does a utility function exist?" instantly:

- A represented preference is always *complete* (real numbers are always comparable).
- A represented preference is always *transitive* ($\geq$ on reals is transitive).

So if either property fails, the answer is "no, because…"

How to check a weird preference (last year's 1.1–1.3)

1. **Completeness:** look for a pair that *no* rule ranks. One counterexample kills it. Try bundles where one coordinate is bigger and the other is smaller, e.g. $(1,0)$ vs. $(0,1)$.
2. **Transitivity:** look for a cycle, or for "small steps that add up" (threshold rules).
3. **Utility:** fails if either of the above fails. If both hold, check continuity on infinite sets.

Drills (answers right below each)

(a) $x \succsim y \iff x_1 \geq y_1$ and $x_2 \geq y_2$ on $[0,1]^2$.
Not complete: $(1,0)$ and $(0,1)$ are unranked. *Transitive:* the inequalities chain. *No utility*, because it is not complete (even though it is continuous).
(b) $x \succsim y \iff x_1 + x_2 \geq y_1 + y_2 - 1$ on $\mathbb{R}^2_+$.
Complete: if $s_x \geq s_y$ then $x \succsim y$, otherwise $y \succsim x$. *Not transitive:* $(0,0) \succsim (0.8,0) \succsim (1.6,0)$ but $(0,0) \not\succsim (1.6,0)$. *No utility.*
(c) Last year's 1.1–1.3. The rules say $x \succ x'$ if $x_1 > x'_1$ and $x_1 + x_2 > x'_1 + x'_2$, *and also* $x' \succ x$ under the identical condition (as printed), plus $x \succ x'$ if $x_1 = x'_1$ and $x_2 < x'_2$.
Points to make: pairs like $x = (1,0)$, $x' = (0,1)$ ($x_1 > x'_1$ but equal sums) are ranked by no rule, so the preferences are *not complete*. The first two lines as printed give both $x \succ x'$ and $x' \succ x$ for the same pair, which contradicts the definition of strict preference. Say so explicitly. Either way there is *no utility function*. When an exam relation looks contradictory, the grader wants you to notice it.

3 Expected utility and independence (20 min)

Definitions

- A **simple lottery** is a probability vector $L = (p_1, \ldots, p_K)$ over outcomes $z_1, \ldots, z_K$. A **compound lottery** randomizes over lotteries. We reduce it by multiplying probabilities.
- **Expected utility (vNM):** there is a utility index u on outcomes such that $L \succsim L' \iff \sum_k p_k u(z_k) \geq \sum_k p'_k u(z_k)$.
- **Independence axiom:** for all L, L', L'' and $\alpha \in (0, 1)$:

$$L \succsim L' \iff \alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''.$$

- **vNM theorem:** complete + transitive + continuous + **independent** $\Rightarrow$ EU representation.

The idea in one line: mixing both options with the *same* third lottery should not change which one you prefer. The common part is irrelevant. (Notes' example: if \$10 for sure $\succ$ \$0 for sure, then a 50/50 mix of \$10 and \$5 must beat a 50/50 mix of \$0 and \$5.)

Why it is *exactly* the assumption EU needs (last year's 3.1, so write this out once from memory):

1. Completeness, transitivity and continuity already give *some* utility $U(L)$ over lotteries, but U could be any nonlinear function of the probabilities.
2. EU means $U(L) = \sum p_k u(z_k)$, which is **linear in probabilities**. Linear implies $U(\alpha L + (1-\alpha)L'') = \alpha U(L) + (1-\alpha)U(L'')$.
3. Independence is precisely the preference-level statement of that linearity: the ranking depends only on the parts that differ. Geometrically, indifference curves in the probability simplex are straight, parallel lines.
4. So it is the one additional condition that turns "some utility over lotteries" into "expected utility." The Allais paradox is the classic violation.

Allais (a violation example): people prefer \$1M for sure over {10%: \$5M, 89%: \$1M, 1%: \$0}, yet prefer {10%: \$5M, 90%: \$0} over {11%: \$1M, 89%: \$0}. The two pairs differ only by a common 89% branch, so independence says the ranking cannot flip.

4 Game theory (60 min)

4.1 The normal form (guaranteed question)

Normal form

A normal-form game is $G = (N, \{S_i\}_{i \in N}, \{u_i\}_{i \in N})$ where

- $N = \{1, \ldots, n\}$ is the set of **players**;
- S_i is player i's set of **strategies**;
- $u_i : S \to \mathbb{R}$, with $S = \prod_j S_j$, is player i's **payoff** for every strategy profile.

All the fundamental solution concepts are defined on the normal form.

Strategies vs. actions (the key idea):

- **Simultaneous game:** strategies = actions.
- **Sequential game:** a strategy is a **complete contingent plan**, an action at *every* node where you might move, *including nodes that are never reached*.
- If player 2 observes player 1's m possible actions and has k actions herself, she has k^m strategies. With 2 and 2 that is 4; with 2 and 3 (PS2) it is $3^2 = 9$.
- The normal forms of the simultaneous and sequential versions differ *only* in the strategy sets (PS2 #3).

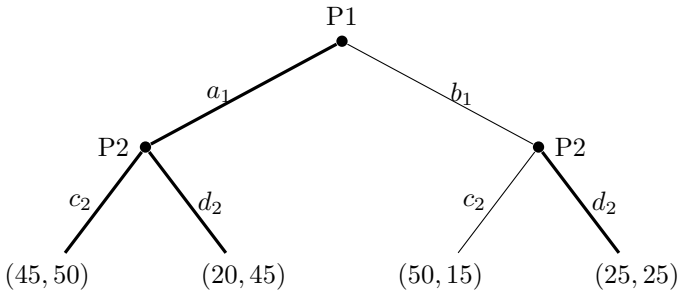
Recipe: a 2-stage sequential game (last year's 2.1–2.5)

Payoffs $u_1(a_1, c_2) = 45$, $u_2 = 50$; $u_1(a_1, d_2) = 20$, $u_2 = 45$; $u_1(b_1, c_2) = 50$, $u_2 = 15$; $u_1(b_1, d_2) = 25$, $u_2 = 25$. Player 1 moves first, and player 2 observes.
Step 1, normal form: $N = \{1, 2\}$, $S_1 = \{a_1, b_1\}$, $S_2 = \{c_2c_2, c_2d_2, d_2c_2, d_2d_2\}$, where xy means "x after a_1, y after b_1."
Step 2, matrix: each cell is the outcome actually reached. For the row a_1, read the *first* letter of P2's strategy; for the row b_1, read the *second*.

	c_2c_2	c_2d_2	d_2c_2	d_2d_2
a_1	45, 50	45, 50	20, 45	20, 45
b_1	50, 15	25, 25	50, 15	25, 25

Step 3, IDSDS: P1 compares $a_1 = (45, 45, 20, 20)$ with $b_1 = (50, 25, 50, 25)$; there is no dominance. P2's payoffs against (a_1, b_1) are $c_2c_2 = (50, 15)$, $c_2d_2 = (50, 25)$, $d_2c_2 = (45, 15)$, $d_2d_2 = (45, 25)$. c_2d_2 strictly beats d_2c_2, so delete it. All other comparisons involve ties (weak dominance only). Next round, P1 compares $a_1 = (45, 45, 20)$ with $b_1 = (50, 25, 25)$: still nothing. Survivors: $\{a_1, b_1\} \times \{c_2c_2, c_2d_2, d_2d_2\}$.
Step 4, NE: mark best responses. P1: vs $c_2c_2 \to b_1$, vs $c_2d_2 \to a_1$, vs $d_2d_2 \to b_1$, vs $d_2c_2 \to b_1$. P2: vs $a_1 \to \{c_2c_2, c_2d_2\}$, vs $b_1 \to \{c_2d_2, d_2d_2\}$. The cells where both are best responding: $NE = \{(a_1, c_2d_2),\ (b_1, d_2d_2)\}$.
Step 5, SPE (backward induction on the tree): after a_1, P2 picks c_2 ($50 > 45$); after b_1, P2 picks d_2 ($25 > 15$). P1 compares 45 with 25 and picks a_1. $SPE = (a_1, c_2d_2)$. The other NE, (b_1, d_2d_2), rests on an **incredible threat**: "I'll play d_2 if you choose a_1," which

P2 would never carry out ($45 < 50$).



Thick lines are the backward-induction choices.

4.2 Solution concepts

Definitions

- **Strictly dominated:** s_i is strictly dominated by s_i' if $u_i(s_i', s_{-i}) > u_i(s_i, s_{-i})$ for *every* s_{-i}.
- **Strictly dominant:** s_i^* strictly dominates every other strategy in S_i.
- **IDSDS:** start from S_i and repeatedly delete strategies that are strictly dominated given the strategies *still remaining* for the others.
- **Rationalizable:** survives repeated removal of strategies that are not a best response to *any* belief about the others' remaining strategies.
- **Best response:** $BR_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i})$.
- **Nash equilibrium:** s^* such that $s_i^* \in BR_i(s_{-i}^*)$ for every i. Equivalently, $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$ for all s_i and all i: no profitable *unilateral* deviation.
- **SPE:** a NE that also induces a NE in every subgame. In finite games of perfect information, use backward induction.

The hierarchy (what each assumes):

Concept	Assumes	Use
Dominant strategies	rationality only	design games so everyone has one (robust)
IDSDS / rationalizability	+ common knowledge of rationality	"RAT brutal": weak predictions
Nash equilibrium	+ correct beliefs about others' play	sharp; "NE easy"

NE always sits inside the set of rationalizable outcomes (NE $\subseteq$ survivors of IDSDS). The reverse does not hold.

Exam tip

- Only **strict** dominance deletes. If two strategies tie in any column, it is not strict, so do not delete. Sequential-game matrices are full of ties, because P2's plan at an unreached node does not change the payoff.
- To find pure NE: underline each player's best response in every row/column. Cells with both underlined are the NE.
- NE of a sequential game that are *not* SPE rest on **incredible threats** (a plan P2 would never carry out at that node).

PS2 #1: IDSDS and NE (Greg rows, Phil columns)

	n_P	l_P	h_P
n_G	0, 0	1, 2	1, 2
l_G	3, 1	3, 3	3, 4
h_G	2, 1	4, 3	5, 5

Delete n_G (dominated by l_G: $3 > 0, 3 > 1, 3 > 1$). Then delete n_P (dominated by l_P: $3 > 1, 3 > 1$). Then delete l_G (dominated by h_G: $4 > 3, 5 > 3$). Then delete l_P (dominated by h_P: $5 > 3$). Survivor: (h_G, h_P), which is also the unique NE.

PS2 #2: a 2×3 sequential game (P2 has 9 strategies)

P1 picks a_1/b_1, then P2 observes and picks $c_2/d_2/e_2$. The payoffs are $(a, c) = (10, 8)$, $(a, d) = (8, 9)$, $(a, e) = (12, 11)$, $(b, c) = (13, 0)$, $(b, d) = (1, 10)$, $(b, e) = (10, 6)$.
$S_2 = \{cc, cd, ce, dc, dd, de, ec, ed, ee\}$ (first letter after a_1, second after b_1).
P2's best responses: vs a_1, any strategy starting with e; vs b_1, any strategy ending in d.
NE with a_1: P2 $\in \{ec, ed, ee\}$, and P1 must prefer a_1 (12) to b_1. Against ec, b_1 gives 13, so that fails. Against ed (b_1 gives 1) and ee (b_1 gives 10) it works.
NE with b_1: P2 $\in \{cd, dd, ed\}$, and b_1 gives 1, which is worse than a_1 every time. None.
$NE = \{(a_1, ed), (a_1, ee)\}$. Backward induction: e after a_1, d after b_1, P1 picks a_1. $SPE = (a_1, ed)$.
(a_1, ee) uses an incredible threat: e after b_1, where P2 would really pick d ($10 > 6$).
(The printed official key for this question reads "$(b; ec)$", which looks garbled. Check that the logic above matches your own.)

4.3 Mixed strategies

Mixed strategies

A **mixed strategy** $\sigma_i \in \Delta(S_i)$ is a probability distribution over pure strategies. A **mixed NE** is σ^* with $U_i(\sigma_i^*, \sigma_{-i}^*) \geq U_i(\sigma_i, \sigma_{-i}^*)$ for all σ_i and all i. **Every finite game has at least one NE in mixed strategies.**

Recipe (the indifference trick): if a player mixes, she must be *indifferent* between the pure strategies she uses. So choose *my* probability to make *the other player* indifferent.

Matching pennies: P1 wins on a match. With $q = \Pr(\text{P2 plays } H)$: $U_1(H) = 2q - 1$ and

$U_1(T) = 1 - 2q$, so indifference gives $q = 1/2$. By symmetry $p = 1/2$. The mixed NE is $(1/2, 1/2)$.

5 Games where money is not the payoff

This is last year's Section 2.2 format, and PS2 #4 is the idea behind it.

The key point: the money matrix is *not* the game matrix. Payoffs come from *preferences*. Before solving, convert money into each player's utility for each case.

Last year's money game: $(R, R) = \$9, \$9$; $(R, Q) = \$0, \$20$; $(Q, R) = \$20, \$0$; $(Q, Q) = \$2, \$2$; sequential

Case 1 (selfish): after R_1, P2 plays Q ($20 > 9$); after Q_1, P2 plays Q ($2 > 0$). P1 gets 0 from R and 2 from Q, so she plays Q. $SPE = (Q_1, Q_2Q_2)$ with outcome (\$2, \$2), a Prisoner's Dilemma. NE: in the 2×4 matrix, only (Q_1, Q_2Q_2).
Case 2 (each maximizes the total): totals are $RR = 18$, $RQ = 20$, $QR = 20$, $QQ = 4$. After R_1, P2 plays Q ($20 > 18$); after Q_1, P2 plays R ($20 > 4$). P1 is indifferent ($20 = 20$). SPE: (R_1, Q_2R_2) and (Q_1, Q_2R_2).
Case 3 (Nature picks 2a with probability μ or 2b; P1 cares about her money + 2b's): vs 2a (selfish), 2a plays Q always, so P1 gets 0 after R_1 and 2 after Q_1. vs 2b (cares about the sum with P1), P1's payoff is the total: 20 after either action. $E[R_1] = 20(1 - \mu)$ and $E[Q_1] = 2\mu + 20(1 - \mu)$, so Q_1 **is strictly better for every** $\mu \in (0, 1)$.
Difficulties to mention: P1 doesn't know which subgame she is in (imperfect information), so we need a Nature move, an information set, beliefs μ and expected payoffs. That goes beyond plain backward induction.

PS2 #4 lesson ("third-party observation"): seeing everyone play M in a money game does *not* reject game theory or rationality. We don't know players' preferences over money, so we can always build payoffs that make (M_1, M_2) the unique NE. To test rationality you need to know preferences.

6 Cournot oligopoly from scratch (75 min with §6)

6.1 The story

Firms produce a **homogeneous** good and **choose quantities** at the same time. Total output Q then hits the market, and the price adjusts to clear it: $p = P(Q)$. Think of bottled-water suppliers who must commit to output before they know the price. Each firm's best quantity depends on what the others produce, which makes it a game.

Notation: q_i is firm i's output, $Q = \sum_j q_j$, and $Q_{-i} = Q - q_i$ is everyone else's output. Inverse demand is $P(Q)$ and the cost is $C_i(q_i)$.

6.2 Normal form of a Cournot game (say this every time)

$$N = \{1, \ldots, n\}, \qquad S_i = [0, \bar{q}] \text{ (quantities)}, \qquad \pi_i(q_i, Q_{-i}) = \underbrace{P(q_i + Q_{-i})\, q_i}_{\text{revenue}} - \underbrace{C_i(q_i)}_{\text{cost}}.$$

6.3 Calculus you need (only this)

- $\frac{d}{dq}(aq) = a$, $\frac{d}{dq}(q^2) = 2q$, $\frac{d}{dq}(\frac{c}{2}q^2) = cq$. Anything that doesn't contain q_i (like Q_{-i}) is a *constant* when you differentiate with respect to q_i.
- **FOC:** set $\partial\pi_i/\partial q_i = 0$ and solve for q_i.
- **SOC:** $\partial^2\pi_i/\partial q_i^2 < 0$ means profit is concave, so the FOC gives a maximum. With linear demand and linear or quadratic costs it always holds; say so in one line.

6.4 The 4-step recipe

Solving any Cournot game

Step 1. Profit. Write $\pi_i = (a - q_i - Q_{-i})q_i - C(q_i)$ *before* imposing anything.

Step 2. Best response. Differentiate with respect to q_i only, set it to 0, and solve: $q_i = b_i(Q_{-i})$. It slopes down, because quantities are *strategic substitutes*. Truncate at 0 if negative.

Step 3. Equilibrium. NE means every firm is on its BR at once. With identical firms, *now* impose symmetry: $q_i = q^c$ and $Q_{-i} = (n-1)q^c$, and solve the single equation. With different firms, solve the BR equations together.

Step 4. Everything else. $Q^c = nq^c$, $p^c = P(Q^c)$, π_i, CS, PS, TS, the efficient outcome, DWL, and the graph.

Exam tip

The notes' #1 mistake: **do not set $q_1 = q_2$ before differentiating.** A firm controls only its own q_i and takes Q_{-i} as given. Impose symmetry only in Step 3. Second mistake: treating Q as if one firm controls all of it.

6.5 Worked: the lecture example (from your 10/22 notes)

$P(Q) = 100 - Q$, $c(q_i) = 10q_i$, $n = 3$.

1. $\pi_i = (100 - q_i - Q_{-i})q_i - 10q_i$.
2. FOC: $100 - 2q_i - Q_{-i} - 10 = 0 \Rightarrow b_i(Q_{-i}) = \dfrac{90 - Q_{-i}}{2}$.
3. Symmetry, $Q_{-i} = 2q^c$: $q^c = \dfrac{90 - 2q^c}{2} \Rightarrow 4q^c = 90 \Rightarrow q^c = 22.5$.
4. $Q^c = 67.5$, $p^c = 32.5$, $\pi_i = (32.5 - 10)(22.5) = 506.25$. Efficient: $p = MC = 10 \Rightarrow Q^* = 90$. $CS = \frac{1}{2}(67.5)^2 = 2278.1$, $PS = 1518.75$, $DWL = \frac{1}{2}(90 - 67.5)(32.5 - 10) = 253.1$.

(Your notes wrote "$q^c = 90/4$"; that is exactly 22.5. Notes check: correct.)

6.6 Welfare: what to compute and draw

Welfare definitions

- **Consumer surplus** CS: the area under demand and above p^c, up to Q^c. With $P = a - Q$ it is $\frac{1}{2}Q^c(a - p^c) = \frac{1}{2}(Q^c)^2$.
- **Producer surplus** PS: the area above MC and below p^c, up to Q^c. With no fixed costs, $PS = \sum_i \pi_i$.
- **Total surplus** $TS = CS + PS$, the total gains from trade.
- **Efficient outcome:** maximize TS, i.e. produce until $P(Q) = MC$, using the cheapest producers. If each firm has rising MC, split output so their MCs are equal.
- **Deadweight loss** $DWL = TS^* - TS^c$: the surplus lost from trades worth more to buyers than they cost to make that don't happen. Graphically, the triangle between demand and MC from Q^c to Q^*.

Why Cournot is inefficient: when firm i expands, the price falls on *all* units, including rivals'. Firm i counts only its own loss ($P'(Q)q_i$), so each firm restricts output to keep the price up, the market **underproduces**, and $p > MC$. This is a NE that is not Pareto efficient, just like the Prisoner's Dilemma. Firms "make more gains from trade themselves" by selling less at a high price.

Benchmarks to remember (linear demand $a - bQ$, constant MC c): $q^c = \dfrac{a-c}{(n+1)b}$, $p^c = \dfrac{a+nc}{n+1}$. $n = 1$ is monopoly; as $n \to \infty$, $p \to c$ (competitive). Ordering: $Q^m < Q_2^c < Q_3^c < \cdots < Q^{PC}$ and $p^m > p_2^c > \cdots > MC$. The Cournot markup is $\dfrac{p - MC_i}{p} = \dfrac{s_i}{|\varepsilon_D|}$ (a bigger market share or less elastic demand means a bigger markup).

7 Problem Set 3, fully solved (Cournot parts)

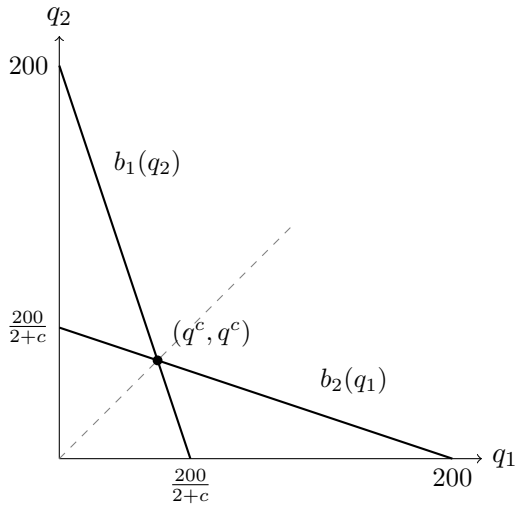
Try each part yourself first, then check. PS3 #4 is Bertrand, which the professor said is not on the midterm, so it is skipped here.

PS3 #1: Cournot duopoly, $P(Q) = 200 - Q$, $C(q_i) = cq_i^2/2$

1.a Profit. $\pi_i = (200 - q_i - q_j)q_i - \dfrac{c}{2}q_i^2$.

1.b Best response. $\dfrac{\partial \pi_i}{\partial q_i} = 200 - 2q_i - q_j - cq_i = 0 \Rightarrow \boxed{b_i(q_j) = \dfrac{200 - q_j}{2 + c}}$. SOC: $-(2 + c) < 0$.

Graph: b_1 hits the q_1-axis at $200/(2 + c)$ and the q_2-axis at 200. b_2 is its mirror image. They cross on the 45° line.

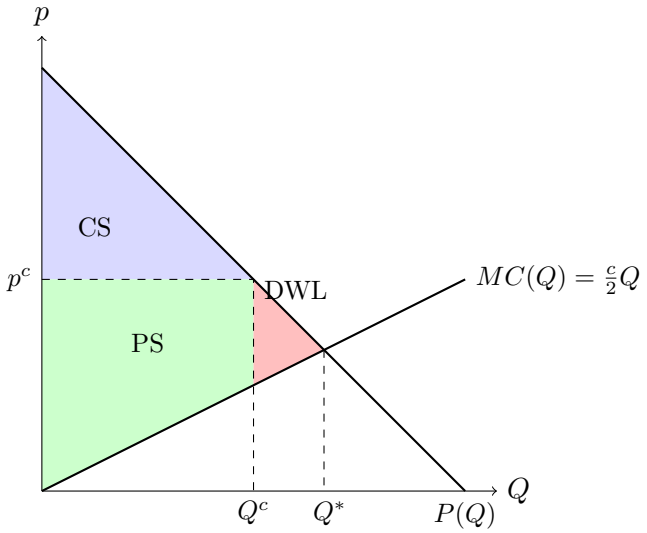


Drawn for $c = 1$: the intercepts are 66.7 and the equilibrium is $(50, 50)$.

1.c Equilibrium. With symmetry,

$$(2+c)q = 200 - q \;\Rightarrow\; \boxed{q^c = \frac{200}{3+c}}, \qquad Q^c = \frac{400}{3+c}, \qquad \boxed{p^c = 200 - \frac{400}{3+c} = \frac{200(1+c)}{3+c}}.$$

1.d Graph on the p–Q plane. Demand $P = 200 - Q$. The industry MC curve: each firm has $MC = cq_i$, and with output split evenly ($q_i = Q/2$), $MC(Q) = \frac{c}{2}Q$. Mark (Q^c, p^c) on demand, the efficient point where demand crosses $MC(Q)$, and shade CS, PS, DWL.



Drawn for $c = 1$: $Q^c = 100$, $p^c = 100$, $Q^* = 133.3$, $p^* = 66.7$.

1.e Profits and welfare.

$$\pi_i = p^c q^c - \tfrac{c}{2}(q^c)^2 = q^c\Big(p^c - \tfrac{c}{2}q^c\Big) = \frac{200}{3+c}\cdot\frac{200(1+c)-100c}{3+c} = \boxed{\frac{20000(2+c)}{(3+c)^2}}$$

$$CS = \tfrac{1}{2}Q^c(200-p^c) = \tfrac{1}{2}(Q^c)^2 = \boxed{\frac{80000}{(3+c)^2}}, \qquad PS = 2\pi_i = \frac{40000(2+c)}{(3+c)^2}$$

$$TS = CS + PS = \boxed{\frac{40000(4+c)}{(3+c)^2}}$$

Efficient: $200 - Q = \frac{c}{2}Q \Rightarrow Q^* = \dfrac{400}{2+c}$ and $p^* = \dfrac{200c}{2+c}$, so

$$TS^* = 200Q^* - \tfrac{1}{2}(Q^*)^2 - 2\cdot\tfrac{c}{2}\Big(\tfrac{Q^*}{2}\Big)^2 = \frac{40000}{2+c}, \qquad \boxed{DWL = TS^* - TS = \frac{40000}{(2+c)(3+c)^2}}.$$

Check with $c = 1$: $q^c = 50$, $p^c = 100$, $\pi_i = 3750$, $CS = 5000$, $TS = 12{,}500$, $TS^* = 13{,}333$, $DWL = 833$. These match the Ch. 5 notes.

PS3 #2: the same model with n firms

2.a $\pi_i = (200 - q_i - Q_{-i})q_i - \dfrac{c}{2}q_i^2$.

2.b FOC: $200 - 2q_i - Q_{-i} - cq_i = 0 \Rightarrow \boxed{b_i(Q_{-i}) = \dfrac{200 - Q_{-i}}{2+c}}$. On the q_i–Q_{-i} plane this is a line from $q_i = \frac{200}{2+c}$ (at $Q_{-i} = 0$) down to $Q_{-i} = 200$.

2.c Symmetry, $Q_{-i} = (n-1)q$: $(2+c)q = 200 - (n-1)q \Rightarrow \boxed{q^c = \dfrac{200}{n+1+c}}$, $Q^c = \dfrac{200n}{n+1+c}$, $\boxed{p^c = \dfrac{200(1+c)}{n+1+c}}$.

2.d Same picture as 1.d, with the industry MC now $MC(Q) = \frac{c}{n}Q$ (output split n ways), which gets flatter as n grows.

2.e

$$\pi_i = \Big(1 + \frac{c}{2}\Big)\Big(\frac{200}{n+1+c}\Big)^2 = \frac{20000(2+c)}{(n+1+c)^2}, \qquad CS = \tfrac{1}{2}(Q^c)^2 = \frac{20000\,n^2}{(n+1+c)^2},$$

$$PS = n\pi_i = \frac{20000\,n(2+c)}{(n+1+c)^2}, \qquad TS = \frac{20000\,n(n+2+c)}{(n+1+c)^2}.$$

Efficient: $200 - Q = \frac{c}{n}Q \Rightarrow Q^* = \dfrac{200n}{n+c}$, $TS^* = \dfrac{20000\,n}{n+c}$, so

$$\boxed{DWL = \frac{20000\,n}{(n+c)(n+1+c)^2}} \xrightarrow[n\to\infty]{} 0.$$

(For $n = 2$ this reduces to the 1.e formulas, which is a good self-check.) **Interpretation:** more firms means each firm has less effect on price, so the underproduction shrinks and p^c approaches MC. With rising MC, more firms also spread production, which lowers the cost of any given Q.

PS3 #3: differentiated products, $P_1 = 18 - q_1 - \gamma q_2$, $P_2 = 18 - q_2 - \gamma q_1$, no costs, $\gamma \in (0, 1)$

γ measures how substitutable the goods are: $\gamma = 0$ means independent goods, $\gamma = 1$ means identical goods.

- **3.a** $\pi_1 = (18 - q_1 - \gamma q_2)q_1$ and $\pi_2 = (18 - q_2 - \gamma q_1)q_2$.
- **3.b** FOC: $18 - 2q_1 - \gamma q_2 = 0 \Rightarrow b_1(q_2) = \dfrac{18 - \gamma q_2}{2}$; symmetrically $b_2(q_1) = \dfrac{18 - \gamma q_1}{2}$.
- **3.c** Each BR is a line from 9 on its own axis down to $18/\gamma$ on the other axis. A smaller γ makes the lines flatter (less strategic interaction).
- **3.d** Symmetry: $2q = 18 - \gamma q \Rightarrow \boxed{q_i(\gamma) = \dfrac{18}{2+\gamma}}$ and $p_i(\gamma) = 18 - (1+\gamma)\dfrac{18}{2+\gamma} = \boxed{\dfrac{18}{2+\gamma}}$.
  Profit is $\left(\frac{18}{2+\gamma}\right)^2$.
  At $\gamma = 0$ we get two monopolies ($q = p = 9$); at $\gamma = 1$, homogeneous Cournot ($q = p = 6$). More differentiation (a lower γ) gives a higher price and profit, because there is less competition.

8 Market games, efficiency and equity (15 min)

Definitions from the notes

- **Pareto efficient:** there is no way to change production or allocation to make someone better off without hurting anyone else.
- **Productive efficiency:** production happens at minimum cost (you can't make more of one thing without less of another).
- **Allocative efficiency:** each good goes to the person who values it most.
- **Equity:** fairness, which can be defined in many ways. It needs a separate normative criterion.

Points to be able to make:

1. **NE is generally neither efficient nor equitable.** The Prisoner's Dilemma shows inefficiency. Equity is not even part of the definition of NE.
2. **Market games have natural welfare measures:** gains from trade for each person. Buyers get $CS = WTP - p$ and sellers get $PS = p - WTS$. $TS = CS + PS$ is the starting point for evaluation. It measures efficiency, *not* equity, because it ignores who gets the surplus.
3. **Pure exchange (no production):** trade is voluntary, so every trade is mutually beneficial. NE does not need perfect competition, and gains from trade are maximized as long as there are no *transaction costs*. Crime and scams (information problems) are what get in the way.
4. **Adding production** raises two questions: (1) *what* each firm produces (which firms should produce) and (2) *how much* each firm produces (the course focuses on this, using partial equilibrium for one industry). Oligopolies **underproduce** to raise price, and sometimes **the wrong (high-cost) firms produce**.

5. **Asymmetric-cost Cournot example** ($P = 120 - Q$, $c_1 = 20$, $c_2 = 40$): $q_1 = 40$, $q_2 = 20$, $p = 60$. Efficiently, firm 1 would make all 100 units at $p = 20$. $DWL = 1200 = 800$ (underproduction) $+ 400$ (the wrong firm producing 20 units at \$20 extra cost each).

9 Answer templates for the guaranteed questions

"Define the normal form of a game."

A normal-form game is $G = (N, \{S_i\}, \{u_i\})$: a set of players N; for each player a strategy set S_i (in sequential games, a strategy is a complete contingent plan specifying an action at every decision node); and payoff functions $u_i : \prod_j S_j \to \mathbb{R}$ giving each player's payoff at every strategy profile. All the fundamental solution concepts (dominance, rationalizability, NE) are defined on it.

"Define Nash equilibrium."

A profile s^* is a NE if, for every player i and every $s_i \in S_i$, $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$. That is, each player's strategy is a best response to the others' equilibrium strategies, so no one gains by deviating alone.

"What are the solution concepts for the normal form?"

(1) Dominant strategies: only rationality is needed, and it is the goal in game design. (2) Rationalizability / iterated deletion of strictly dominated strategies: adds common knowledge of rationality. (3) Nash equilibrium: adds correct beliefs. NE $\subseteq$ rationalizable. Stronger assumptions give sharper predictions. Finite games always have a NE in mixed strategies.

"Solve this Cournot game."

Normal form $\to$ profit $\to$ FOC/BR (and SOC) $\to$ impose symmetry $\to q^c, Q^c, p^c \to \pi$, CS, PS, TS $\to$ efficient Q^* (from $P = MC$) $\to$ DWL $\to$ graphs (the BR plane and the p–Q plane with CS/PS/DWL shaded) $\to$ one sentence on why it's inefficient.

"Apply game theory to a real conflict: what are the difficulties?" (last year's 3.3)

The normal form requires knowing the players (who exactly? coalitions, factions), the strategy sets (huge, evolving, and often unobservable), and above all the *payoffs*, meaning preferences we cannot observe (ideology, domestic politics, how leaders value lives against territory). There is incomplete and asymmetric information, timing and commitment matter (sequential moves, credibility of threats), the interaction is repeated and dynamic, and players may not be fully rational or may misjudge one another. As PS2 #4 shows, observed behavior can be rationalized by *some* payoffs, so the model has little predictive power without knowledge of preferences.

10 Final checklist (last 5 minutes)

- ☐ I can write the definitions of complete, transitive, representation and continuity from memory.
- ☐ I know: finite + complete + transitive ⇒ utility exists; infinite sets also need continuity; lexicographic preferences are the counterexample.
- ☐ I can state independence and explain why EU needs it (linearity in probabilities).
- ☐ I can write $G = (N, \{S_i\}, \{u_i\})$ and list P2's 4 strategies in a 2×2 sequential game.
- ☐ I can fill the 2×4 matrix (first letter for row 1, second letter for row 2).
- ☐ I delete only *strictly* dominated strategies, and I find NE by underlining best responses.
- ☐ I can do backward induction and spot incredible-threat NE.
- ☐ I can compute a 2×2 mixed NE with the indifference trick.
- ☐ I can convert money into payoffs for "selfish," "cares about the sum," and "unknown type" cases.
- ☐ Cournot: profit → FOC → BR → symmetry *after* the FOC → q^c, p^c → CS/PS/TS → $P = MC$ → DWL → graph.
- ☐ I can explain why Cournot underproduces, and what pure exchange vs. production adds.

After this guide: work the three sample midterms in `Midterm/SampleMidterms/` under exam conditions, then grade with the answer keys.